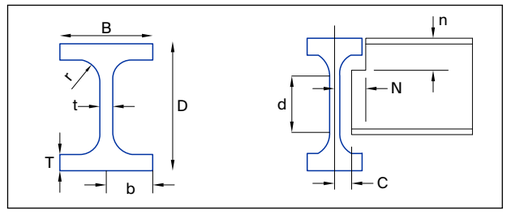
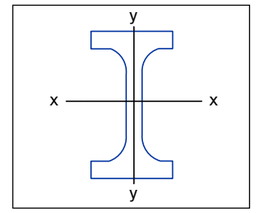
2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W30-30x10 1/2 (762x267)	269	401	825.0	279.1	53.1	29.5	16.5	685.8	511	2.63	23.2	17	135	70
	295	439	835.2	281.7	57.9	32.0	16.5	686.4	559	2.43	21.5	18	135	74
W30-30x15 (762x381)	173	257.5	773.2	380.6	27.1	16.6	16.5	686.1	328	7.04	41.2	10	192	44
	191	284.2	779.3	382.0	30.1	18.0	16.5	686.1	362	6.35	38.1	11	192	47
	211	314	785.9	383.7	33.4	19.7	16.5	686.1	400	5.74	34.9	12	192	50
	235	349.7	795.0	382.4	38.1	21.1	16.5	685.8	445	5.02	32.5	13	191	55
	261	388.4	802.9	384.9	41.9	23.6	16.5	686.1	495	4.59	29.0	14	191	58
	292	434.5	813.1	387.5	47.0	25.9	16.5	686.1	553	4.12	26.5	15	191	63
	326	485.1	823.0	390.4	52.1	29.0	16.5	685.9	617	3.75	23.7	16	191	69
	357	531	833.0	393.0	56.9	31.5	20.0	679.2	677	3.45	21.6	18	191	77
W33-33x11 1/2 (838x292)	118	175.9	834.9	291.7	18.8	14.0	17.8	761.7	224	7.76	54.4	9	149	37
	130	193.8	840.7	292.4	21.7	14.7	17.8	761.7	247	6.74	51.8	9	149	40
	141	209.8	845.8	293.0	24.4	15.4	17.8	761.4	268	6.01	49.5	10	149	42
	152	226.5	850.9	293.8	26.8	16.1	17.8	761.7	289	5.48	47.3	10	149	45
	169	252	859.0	292.1	31.0	17.0	17.8	761.4	319	4.71	44.8	11	148	49
	187	278	865.1	294.1	34.0	19.1	17.8	761.5	355	4.33	39.9	12	148	52
	204	304	871.2	295.7	37.1	20.6	17.8	761.4	386	3.99	37.0	12	148	55
	219	326	877.1	297.2	39.9	22.1	17.8	761.7	416	3.72	34.5	13	148	58
	243	362	885.2	299.7	43.9	24.4	17.8	761.8	460	3.41	31.2	14	148	62
	271	404	895.1	302.1	49.0	26.9	17.8	761.5	513	3.08	28.3	15	148	67
	301	449	905.0	305.2	54.1	30.0	17.8	761.2	572	2.82	25.4	17	148	72
	332	494	915.2	308.1	58.9	33.0	17.8	761.8	629	2.62	23.1	19	148	77
	361	537	925.1	310.6	64.0	35.6	17.8	761.5	684	2.43	21.4	20	148	82
W33-33x15 3/4 (838x400)	201	299.1	855.5	399.9	29.2	18.2	17.8	761.5	381	6.85	41.9	11	201	47
	221	328.9	861.8	401.4	32.4	19.7	17.8	761.4	420	6.20	38.7	12	201	50
	241	358.6	868.2	402.8	35.6	21.1	17.8	761.5	457	5.66	36.1	13	201	53
	263	391.4	877.1	401.4	39.9	22.1	17.8	761.7	499	5.03	34.5	13	200	58
	291	433.1	884.9	404.0	43.9	24.4	17.8	761.4	552	4.60	31.2	14	200	62
	318	473.2	893.1	406.0	48.0	26.4	17.8	761.5	603	4.23	28.8	15	200	66
	354	526.8	903.0	408.9	53.1	29.5	17.8	761.2	672	3.85	25.8	17	200	71
	387	575.9	913.1	411.5	57.9	32.0	17.8	761.7	734	3.55	23.8	18	200	76
	424	631	923.0	414.0	63.0	35.1	20.0	757.0	805	3.29	21.6	20	199	83
	468	697	935.0	418.0	69.1	38.6	20.0	756.8	889	3.02	19.6	21	200	89

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.68	536500	19400	32.4	6.16	13010	1390	15330	2229	0.874	17.0	28.9	3361
2.70	595900	21780	32.7	6.24	14270	1546	16900	2486	0.874	15.8	32.9	4345
3.01	341200	24890	32.3	8.71	8827	1308	9916	2012	0.883	30.3	34.6	635
3.02	381700	28000	32.5	8.80	9796	1466	11030	2258	0.885	27.5	39.3	859
3.04	427000	31500	32.7	8.87	10870	1642	12270	2531	0.886	25.0	44.6	1161
3.05	485900	35570	33.0	8.94	12220	1860	13830	2869	0.889	22.2	50.9	1651
3.07	543700	39920	33.1	8.98	13540	2074	15410	3208	0.888	20.3	57.8	2218
3.10	618300	45680	33.4	9.09	15210	2358	17380	3653	0.889	18.3	67.0	3090
3.12	697600	51790	33.6	9.16	16950	2653	19500	4123	0.888	16.7	77.0	4224
3.14	776600	57770	33.9	9.24	18650	2940	21550	4579	0.888	15.3	87.0	5563
2.78	246000	7800	33.1	5.90	5893	535	6808	842	0.856	46.5	13.0	221
2.79	279200	9067	33.6	6.06	6642	620	7640	974	0.862	41.6	15.2	306
2.80	310200	10250	34.0	6.18	7334	700	8416	1097	0.867	37.9	17.3	404
2.81	339700	11360	34.3	6.27	7985	773	9155	1212	0.870	35.0	19.3	514
2.82	386500	12920	34.8	6.36	8999	884	10310	1383	0.875	30.9	22.1	735
2.84	430400	14470	34.8	6.38	9950	984	11450	1547	0.874	28.3	25.0	983
2.85	473000	16050	35.0	6.45	10860	1086	12530	1710	0.874	26.1	27.9	1265
2.87	513500	17540	35.1	6.49	11710	1180	13550	1864	0.874	24.4	30.7	1569
2.89	573400	19800	35.3	6.56	12960	1321	15060	2095	0.874	22.3	35.0	2095
2.91	648200	22660	35.5	6.64	14480	1500	16910	2385	0.875	20.2	40.5	2884
2.94	729200	25820	35.7	6.72	16120	1692	18920	2704	0.874	18.4	46.7	3910
2.97	810000	28960	35.9	6.79	17700	1880	20890	3018	0.873	17.0	53.1	5088
2.99	892800	32280	36.1	6.87	19300	2078	22880	3346	0.873	15.8	59.8	6494
3.24	479800	31180	35.5	9.05	11220	1559	12640	2405	0.881	31.2	53.2	855
3.26	534700	34970	35.7	9.13	12410	1742	14020	2690	0.882	28.4	60.1	1146
3.27	590000	38800	35.9	9.21	13590	1927	15380	2977	0.884	26.1	67.3	1491
3.29	659000	43070	36.3	9.29	15030	2146	17020	3314	0.888	23.6	75.5	2008
3.31	735400	48390	36.5	9.36	16620	2396	18910	3709	0.887	21.6	85.6	2687
3.33	812500	53680	36.7	9.43	18200	2644	20780	4101	0.888	19.9	95.8	3488
3.35	913500	60680	36.9	9.50	20330	2968	23230	4616	0.887	18.1	110	4737
3.38	1012000	67480	37.1	9.59	22170	3280	25570	5113	0.887	16.7	123	6135
3.40	1120000	74810	37.3	9.64	24260	3614	28140	5652	0.886	15.5	138	7983
3.43	1253000	84520	37.6	9.75	26810	4044	31270	6342	0.886	14.2	158	10569